Software Requirements Specification

for

ResuFlow

AI-Powered CV-to-Portfolio Pipeline

Version 1.0 Approved

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Revision History

Name	Date	Reason For Changes	Version
Team ResuFlow	Feb 28, 2026	Initial document creation	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for ResuFlow, Version 1.0. ResuFlow is an automated pipeline that transforms static CVs into hosted, high-conversion professional portfolios using AI-driven extraction, user-led personalization, and QR-integrated deployment.

This document is intended to serve as the authoritative reference for all development, testing, and deployment activities related to the ResuFlow system. It covers the complete scope of the application from CV ingestion to portfolio hosting.

1.2 Document Conventions

This document follows the IEEE 830-1998 SRS standard. The following conventions apply throughout:

- Requirements are identified using the prefix REQ- followed by a sequential number (e.g., REQ-1).
- Priority levels are indicated as High, Medium, or Low.
- Placeholder items yet to be fully defined are marked [TBD].
- Shall indicates mandatory requirements; should indicates desirable but non-mandatory requirements.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following audiences:

- Developers: Focus on Sections 3, 4, and 5 for functional, interface, and non-functional requirements.
- Testers and QA: Review Section 4 (System Features) and Section 5 (Non-functional Requirements) for acceptance criteria.
- Project Managers: Review Sections 1 and 2 for scope and overall system overview.
- End Users and Stakeholders: Section 2 provides an accessible overview of the product.

Readers are suggested to begin with Section 1 for context and project background, followed by Section 2 for a high-level system description, and then proceed to Sections 3–5 for detailed technical requirements.

1.4 Product Scope

ResuFlow is a zero-code web application that enables non-technical professionals to convert their static CV documents (PDF or DOCX) into fully hosted, personalized, and publicly accessible professional portfolio websites. The system automates AI-driven data extraction, theme-based rendering, cloud hosting, and QR code generation.

The product targets professionals such as marketers, HR personnel, and managers who lack the design or coding skills required to build online portfolios manually. ResuFlow reduces the time from CV submission to live portfolio to a matter of minutes.

The scope includes CV ingestion, AI-powered extraction, user preference configuration, portfolio rendering, automated cloud deployment, and QR code generation. It excludes manual design editing, third-party recruitment platform integrations, and multi-language localization in Version 1.0.

1.5 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
- Gemini API Documentation — Google Developers. https://ai.google.dev/
- React Documentation — https://react.dev/
- Node.js Documentation — https://nodejs.org/
- MongoDB Documentation — https://www.mongodb.com/docs/
- Vercel API Documentation — https://vercel.com/docs/rest-api
- Netlify API Documentation — https://docs.netlify.com/api/get-started/
- qrcode npm package — https://www.npmjs.com/package/qrcode

2. Overall Description

2.1 Product Perspective

ResuFlow is a new, self-contained web application. It does not replace any existing system but fills a gap between standalone resume builders (which produce static PDFs) and full-featured website builders (which require technical knowledge). ResuFlow integrates multiple external services — the Gemini API for AI-based parsing, Vercel/Netlify for hosting, and MongoDB for data persistence — into a unified pipeline.

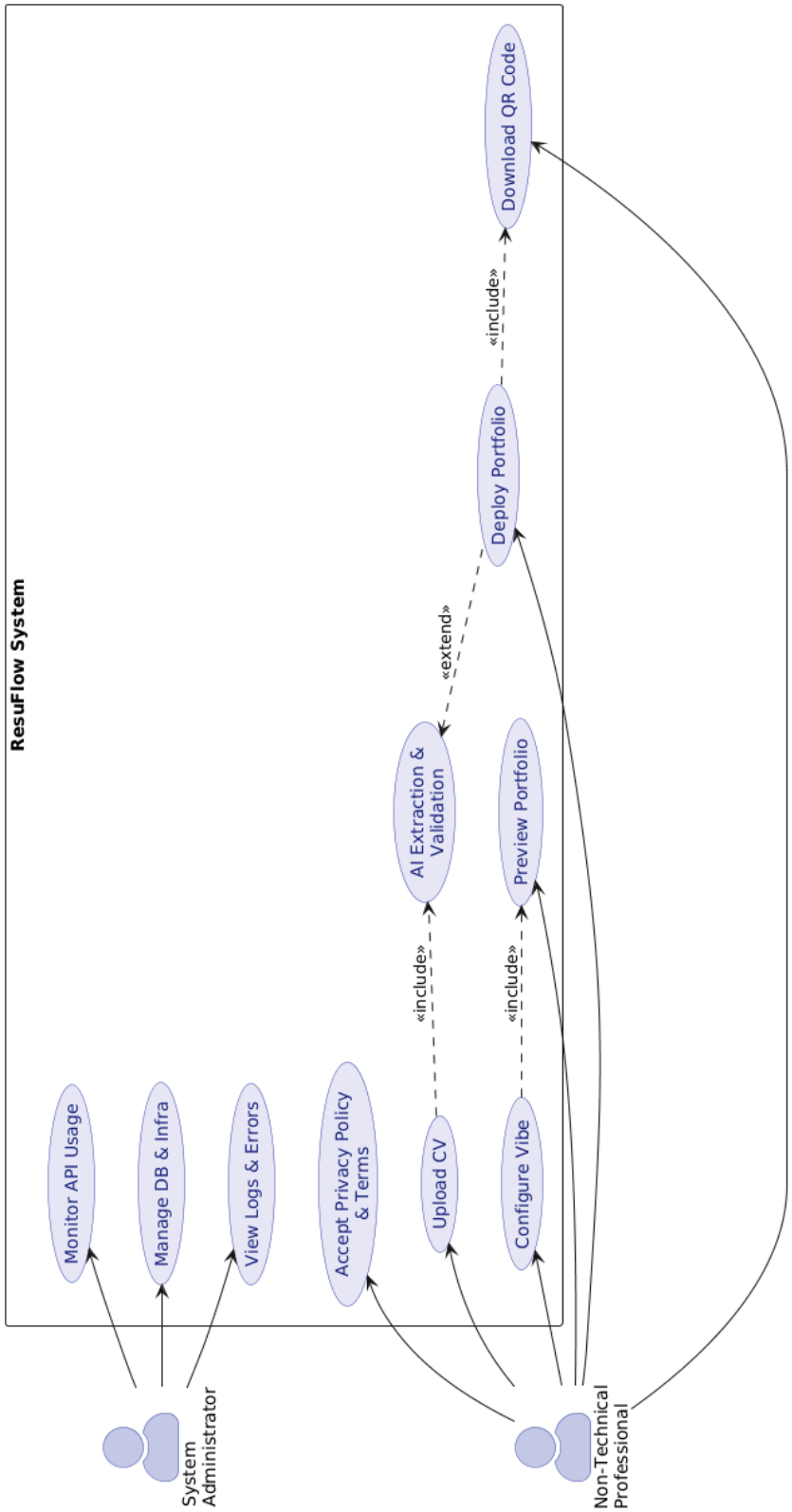
The system architecture follows a client-server model: a React-based frontend communicates with a Node.js/Express backend via REST APIs. The backend orchestrates AI extraction, database operations, theme rendering, and third-party deployments. The system is accessible via any modern web browser and does not require any software installation on the user's device.

2.2 Product Functions

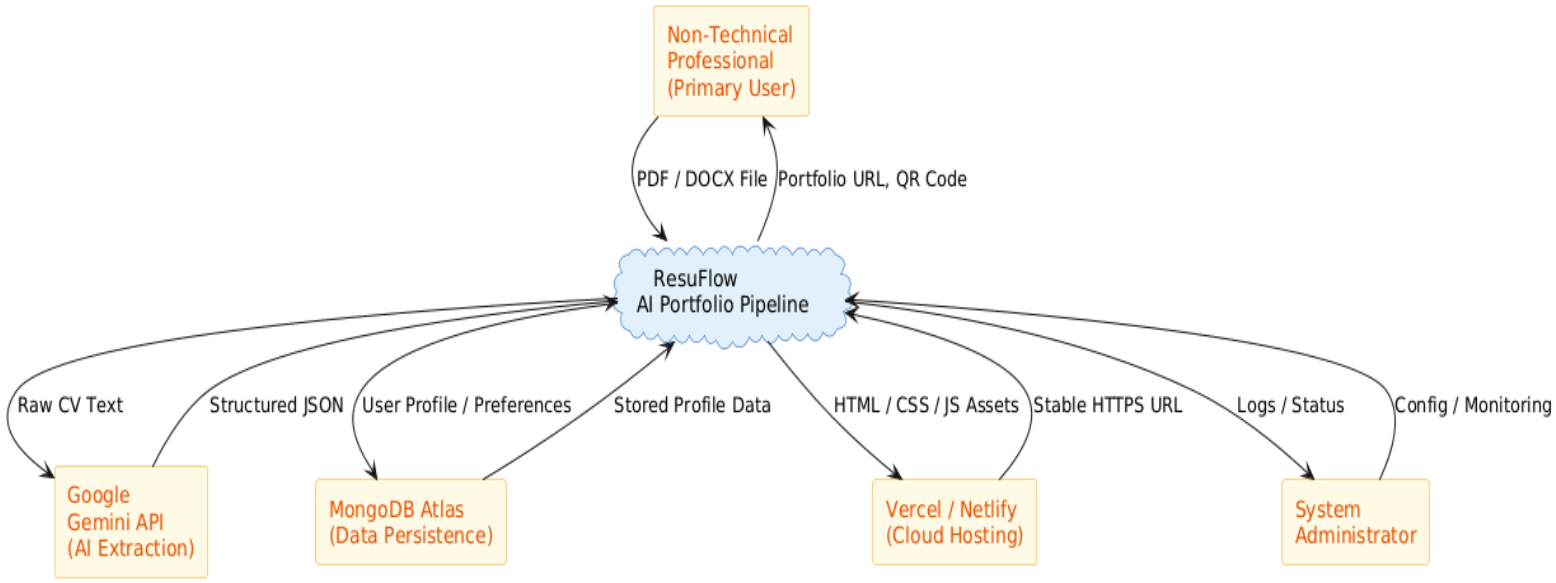
At a high level, ResuFlow provides the following major functions:

- CV Upload and Ingestion: Accept PDF or DOCX files from users and extract raw text content.
- AI-Driven Extraction: Parse unstructured CV text using the Gemini API to produce structured JSON profiles containing skills, work history, education, and achievements.
- Schema Validation: Validate extracted data against a defined JSON schema, ensuring required fields and consistent formats.
- User Preference Configuration: Allow users to select a portfolio "Vibe" (color palette, layout archetype, and About Me tone).
- Dynamic Portfolio Rendering: Generate personalized HTML/CSS portfolio pages from validated JSON profiles and user-selected themes.
- Physical-to-Digital Bridge: Generate QR codes linking physical CVs to the deployed portfolio, enabling instant access for recruiters.

Use Case Diagram



Context Diagram - DFD Level 0



2.3 User Classes and Characteristics

ResuFlow targets two primary user classes:

- Non-Technical Professionals (Primary Users): Marketers, HR managers, project managers, and similar professionals with limited or no coding or design skills. These users need a simple, guided interface. They interact with the CV upload, preference selection, and QR code download features. Their primary concern is ease of use and output quality.
- System Administrators (Secondary Users): Technical users responsible for managing the deployment infrastructure, monitoring API usage (Gemini, Vercel/Netlify), and maintaining the MongoDB database. These users interact with backend dashboards and logs.

2.4 Operating Environment

The ResuFlow system operates in the following environment:

- Frontend: React web application hosted on Vercel/Netlify, accessible from any modern browser (Chrome 110+, Firefox 110+, Safari 15+, Edge 110+).
- Backend: Node.js (v18+) with Express.js, deployed as a serverless or containerized service.
- Database: MongoDB Atlas (cloud-hosted), accessible over HTTPS.
- AI Integration: Google Gemini API (REST-based), accessed from the Node.js backend.
- Hosting APIs: Vercel REST API or Netlify API for automated portfolio deployment.
- Operating System: Platform-independent (cloud-native). No specific OS requirement for end users.

- Network: Requires internet connectivity for all features.

2.5 Design and Implementation Constraints

- The Gemini API is used exclusively for CV parsing; switching to another LLM provider requires significant refactoring of the extraction module.
- Portfolio hosting is limited to Vercel and Netlify free/pro tier API capabilities. Deployment times depend on third-party SLA.
- MongoDB Atlas free tier limits storage to 512 MB, which may constrain scale in production.
- The system must comply with GDPR principles; CV data is stored only for the duration of session processing unless the user explicitly opts to save their profile.
- Frontend must be accessible on devices with a minimum screen width of 360px (mobile-first design).
- The QR code must use Level H error correction (up to 30% data restoration) to ensure scannability on printed media.

2.6 User Documentation

The following user documentation will be delivered with Version 1.0:

- In-App Guided Tour: An interactive onboarding flow shown to first-time users, covering CV upload, vibe selection, and portfolio preview steps.
- FAQ Page: A web-based FAQ addressing common issues such as unsupported CV formats, extraction errors, and QR code usage.
- Video Tutorial: A 3–5 minute walkthrough video hosted on the ResuFlow website demonstrating the end-to-end flow.
- Help Tooltips: Contextual tooltips on all major UI elements explaining their purpose.

2.7 Assumptions and Dependencies

The following assumptions apply to this SRS:

- Users will supply CVs in English; multi-language support is not in scope for v1.0.
- The Gemini API will maintain a stable REST interface compatible with the integration implemented during development.
- Vercel and Netlify will continue to offer programmatic deployment APIs with the current functionality throughout the project lifecycle.
- Users have access to a modern web browser and a stable internet connection.
- CV files submitted by users are well-formed and do not exceed 5 MB in size.
- Third-party hosting services (Vercel/Netlify) are assumed to have 99.9% uptime SLA.

3. External Interface Requirements

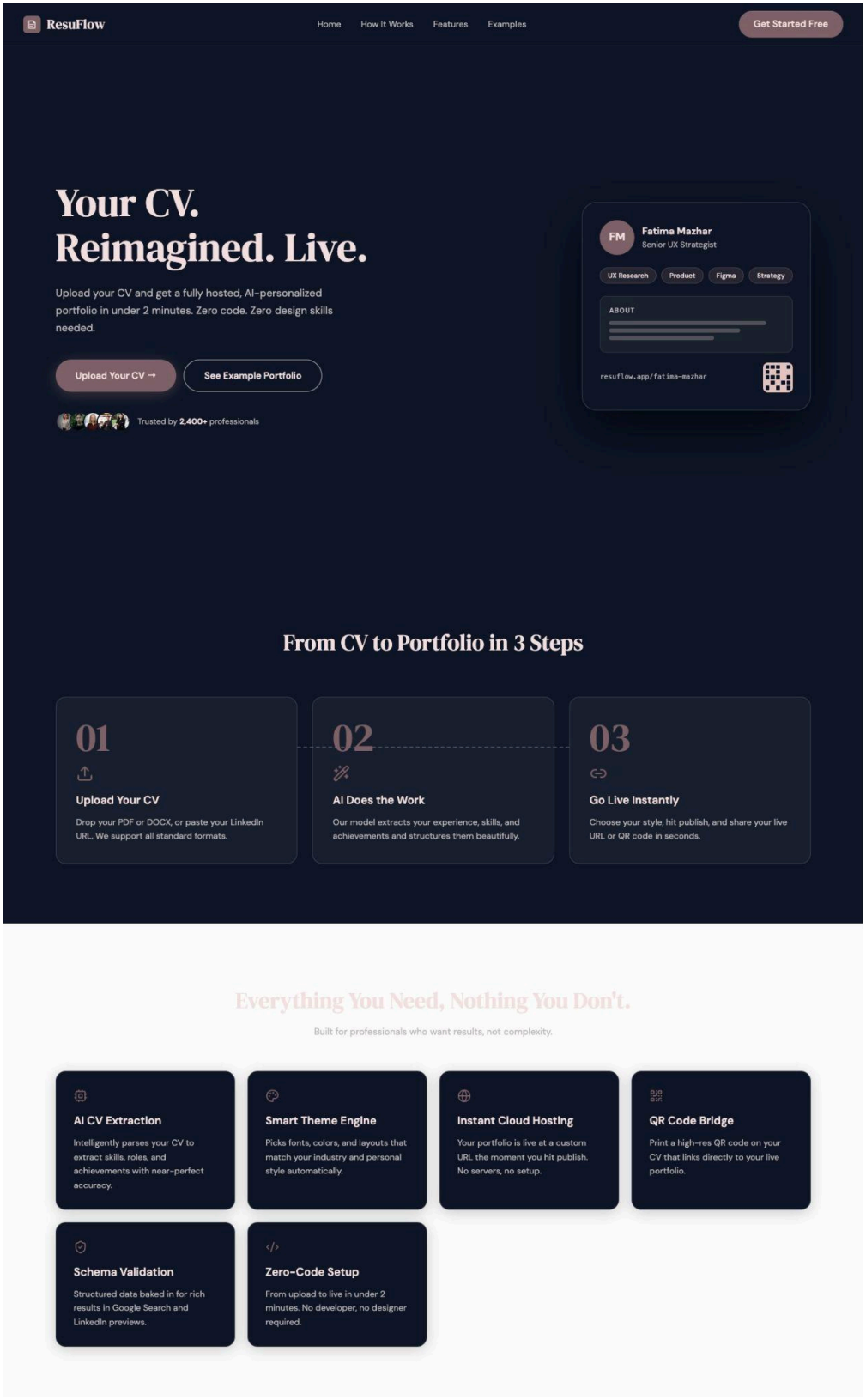
3.1 User Interfaces

The ResuFlow frontend is a React single-page application (SPA) with the following interface screens:

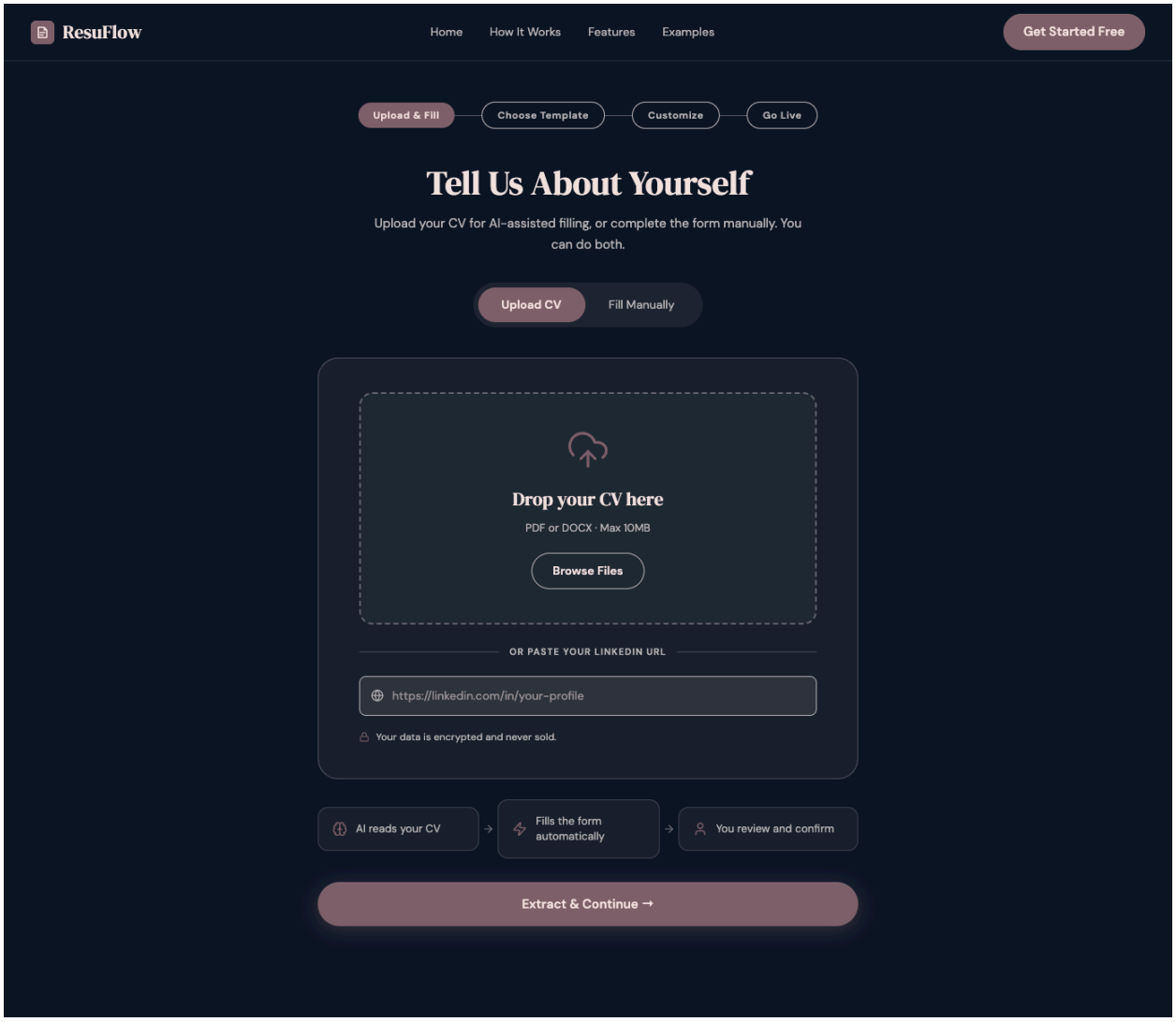
- Landing Page: Marketing page with a clear call-to-action ("Upload Your CV") and brief feature highlights.
- CV Upload Screen: Drag-and-drop file upload area supporting PDF and DOCX formats. Displays file validation feedback (size, format). Shows a progress bar during AI extraction.
- Vibe Selection Screen: Three-panel selector for (1) color palette, (2) layout archetype (e.g., Minimal, Creative, Corporate), and (3) About Me tone (Professional, Casual, Bold). Displays a live thumbnail preview of the selected combination.
- Portfolio Preview Screen: Full-page preview of the generated portfolio. Includes an "Edit Preferences" back button and a "Deploy" action button.
- Dashboard Screen: Post-deployment screen showing the portfolio URL, a download button for the QR code (PNG, 300 DPI), and a copy-link button.

The UI follows a mobile-first responsive design. Minimum supported viewport: 360px wide. Standard WCAG 2.1 AA accessibility guidelines apply.

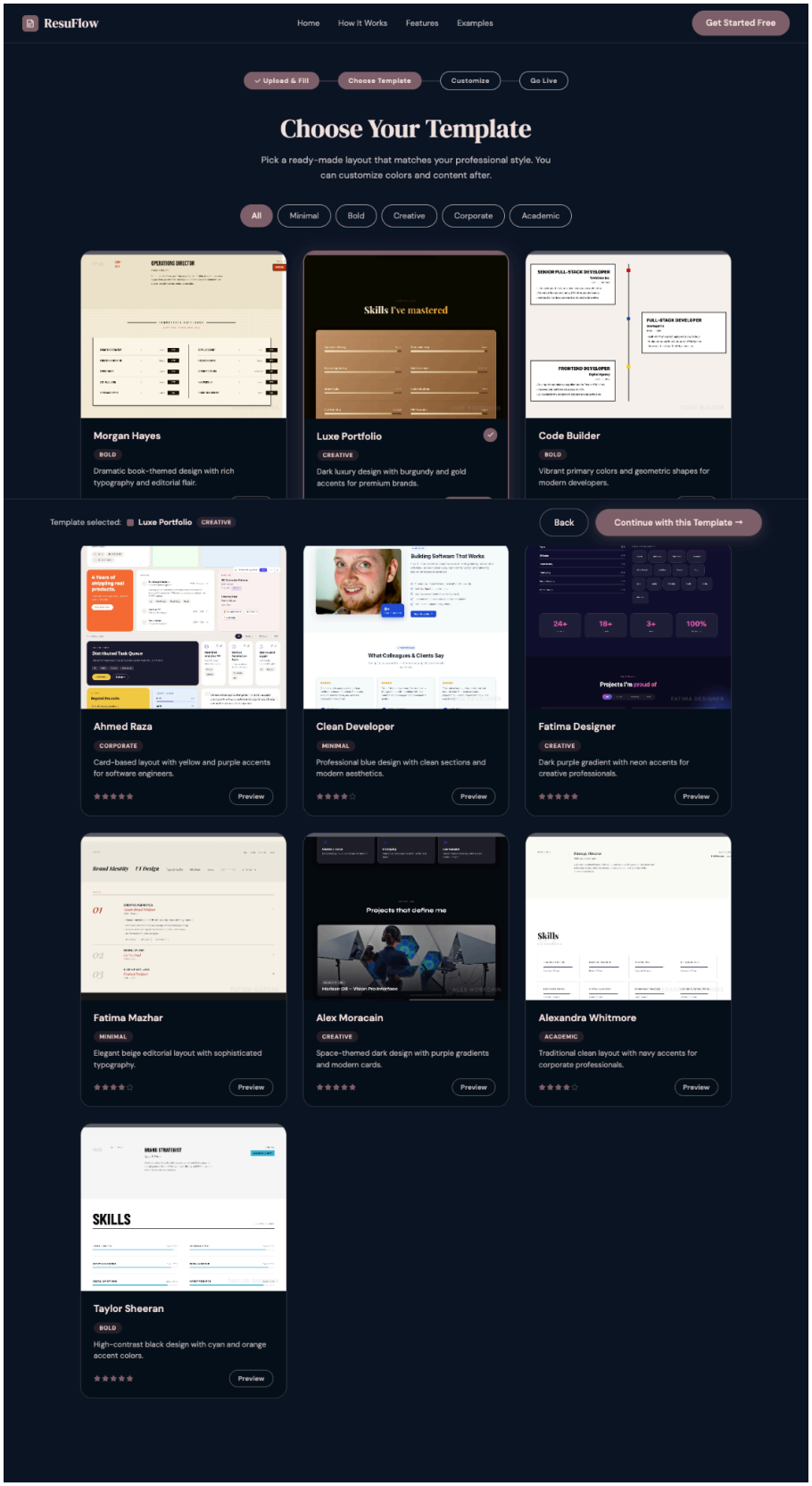
/home


ResuFlow
Home
How It Works
Features
Examples
Get Started Free
Your CV.
Reimagined. Live.
Upload your CV and get a fully hosted, AI-personalized portfolio in under 2 minutes. Zero code. Zero design skills needed.
Upload Your CV →
See Example Portfolio
Trusted by 2,400+ professionals
FM
Fatima Mazhar
Senior UX Strategist
UX Research
Product
Figma
Strategy
ABOUT
resuflow.app/fatima-mazhar
From CV to Portfolio in 3 Steps
01
Upload Your CV
Drop your PDF or DOCX, or paste your LinkedIn URL. We support all standard formats.
02
AI Does the Work
Our model extracts your experience, skills, and achievements and structures them beautifully.
03
Go Live Instantly
Choose your style, hit publish, and share your live URL or QR code in seconds.
Everything You Need, Nothing You Don't.
Built for professionals who want results, not complexity.
AI CV Extraction
Intelligently parses your CV to extract skills, roles, and achievements with near-perfect accuracy.
Smart Theme Engine
Picks fonts, colors, and layouts that match your industry and personal style automatically.
Instant Cloud Hosting
Your portfolio is live at a custom URL the moment you hit publish. No servers, no setup.
QR Code Bridge
Print a high-res QR code on your CV that links directly to your live portfolio.
Schema Validation
Structured data baked in for rich results in Google Search and LinkedIn previews.
Zero-Code Setup
From upload to live in under 2 minutes. No developer, no designer required.

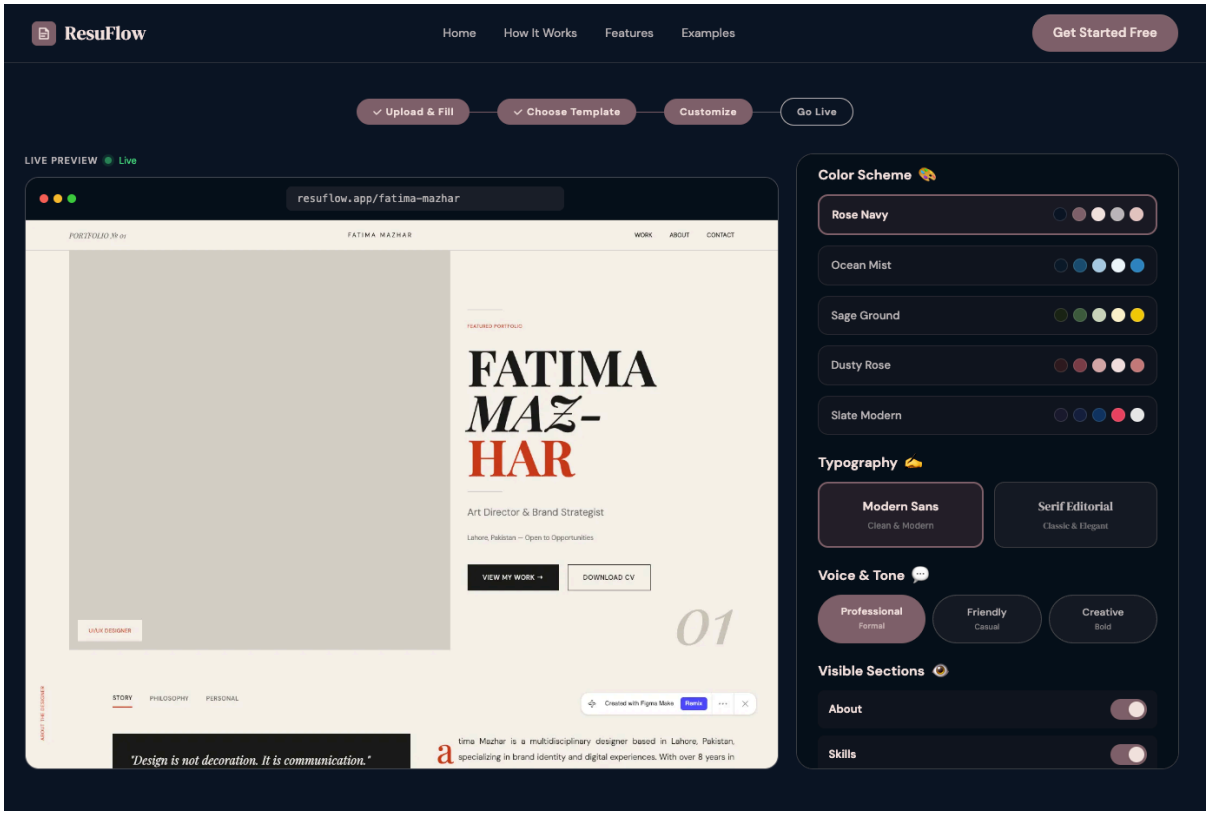
/upload-cv

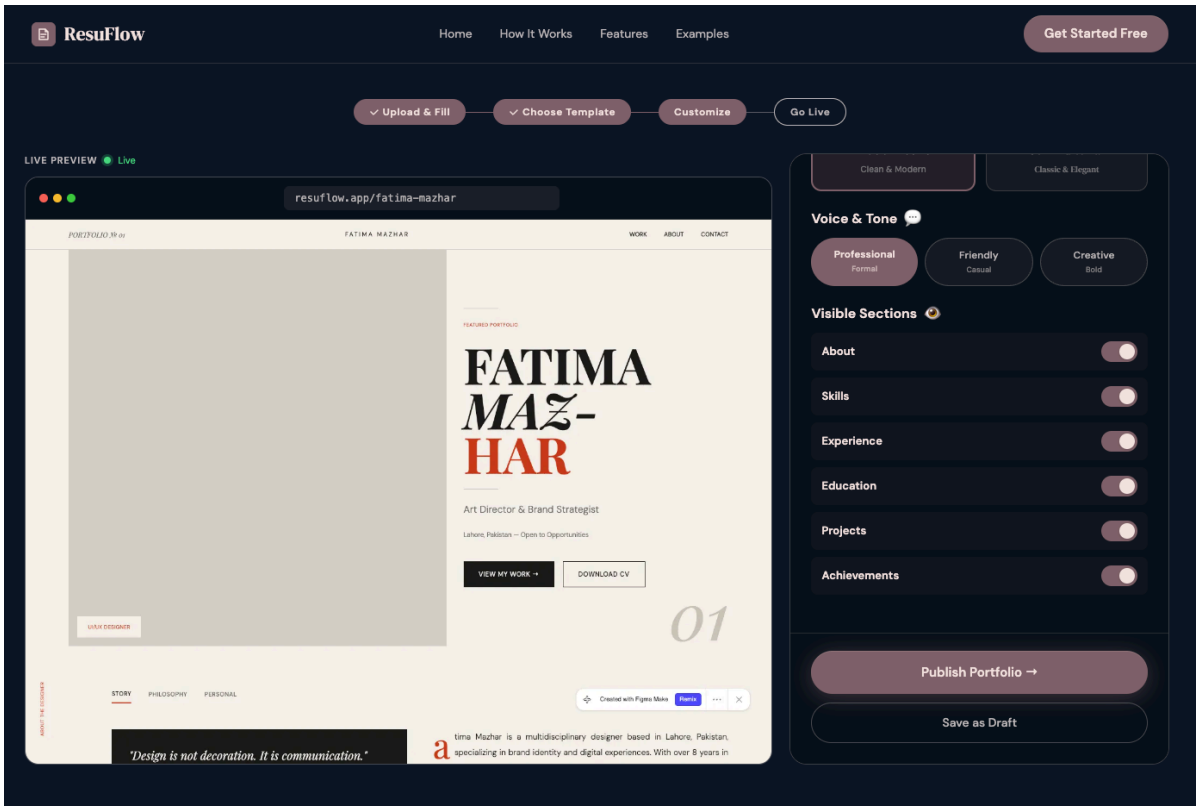

ResuFlow
Home
How It Works
Features
Examples
Get Started Free
Upload & Fill
Choose Template
Customize
Go Live
Tell Us About Yourself
Upload your CV for AI-assisted filling, or complete the form manually. You can do both.
Upload CV
Fill Manually
Drop your CV here
PDF or DOCX · Max 10MB
Browse Files
OR PASTE YOUR LINKEDIN URL
https://linkedin.com/in/your-profile
Your data is encrypted and never sold.
AI reads your CV
Fills the form automatically
You review and confirm
Extract & Continue →

/templates

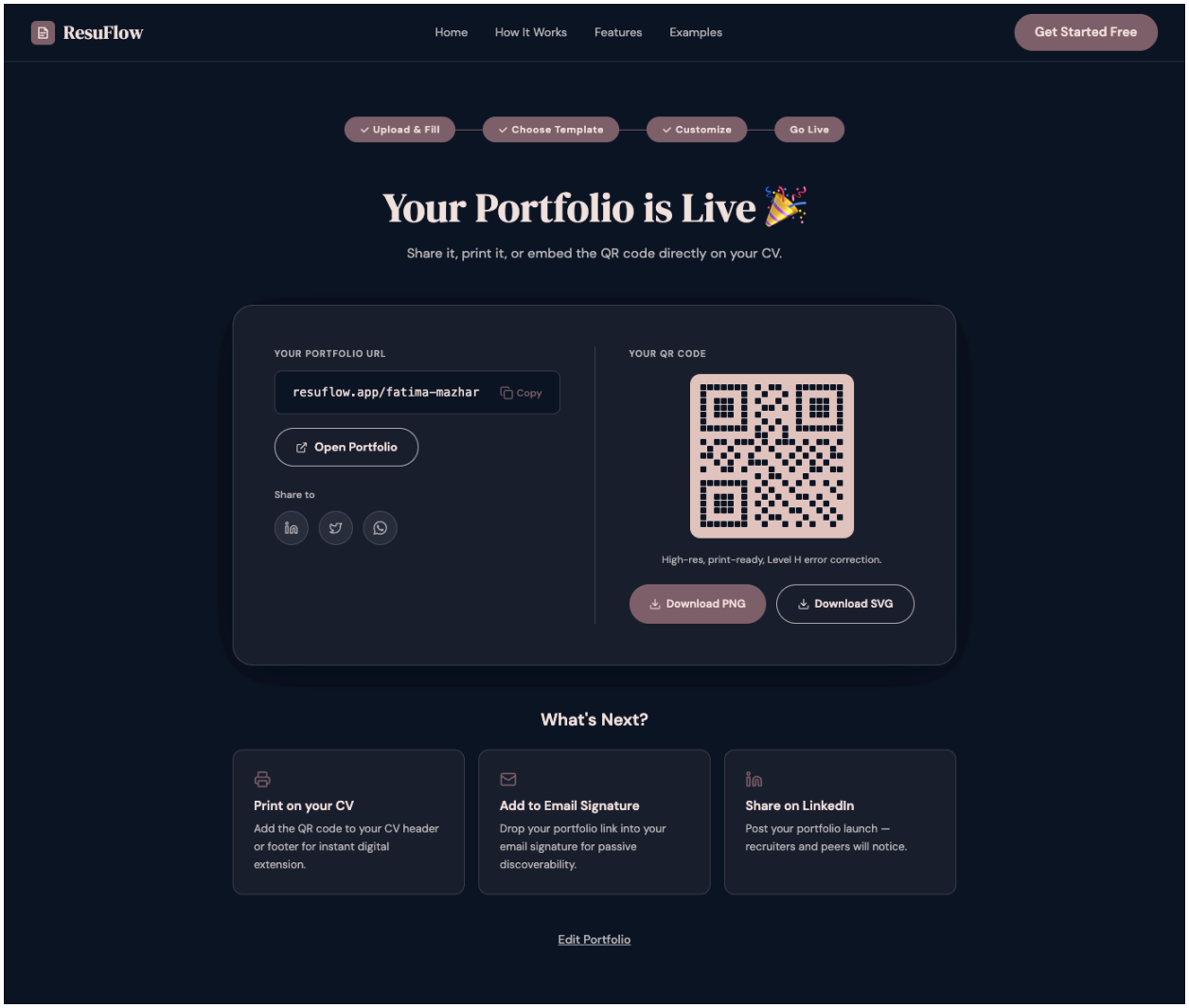

ResuFlow
Home
How It Works
Features
Examples
Get Started Free
Upload & Fill
Choose Template
Customize
Go Live
Choose Your Template
Pick a ready-made layout that matches your professional style. You can customize colors and content after.
All
Minimal
Bold
Creative
Corporate
Academic
Morgan Hayes
BOLD
Dramatic book-themed design with rich typography and editorial flair.
Luxe Portfolio
CREATIVE
Dark luxury design with burgundy and gold accents for premium brands.
Code Builder
BOLD
Vibrant primary colors and geometric shapes for modern developers.
Template selected: Luxe Portfolio CREATIVE
Back
Continue with this Template →
Ahmed Raza
CORPORATE
Card-based layout with yellow and purple accents for software engineers.
Preview
Clean Developer
MINIMAL
Professional blue design with clean sections and modern aesthetics.
Preview
Fatima Designer
CREATIVE
Dark purple gradient with neon accents for creative professionals.
Preview
Fatima Mazhar
MINIMAL
Elegant beige editorial layout with sophisticated typography.
Preview
Alex Moracain
CREATIVE
Space-themed dark design with purple gradients and modern cards.
Preview
Alexandra Whitmore
ACADEMIC
Traditional clean layout with navy accents for corporate professionals.
Preview
SKILLS
Taylor Sheeran
BOLD
High-contrast black design with cyan and orange accent colors.
Preview

/customization





/deploy



3.2 Hardware Interfaces

ResuFlow is a cloud-based web application and does not directly interface with any hardware components. The system relies on the user's device (desktop, tablet, or mobile) to access the web interface via a browser. No specialized hardware is required. QR codes are rendered in software and intended for printing on standard paper or digital display; the system itself has no direct interface with printers.

3.3 Software Interfaces

ResuFlow integrates with the following external software systems:

- Google Gemini API (v1): Used for CV text extraction and structured data generation. Communication via HTTPS REST. Input: raw CV text (string). Output: structured JSON profile. Authentication via API key stored in environment variables.

- MongoDB Atlas: NoSQL database for storing user profiles, extracted JSON data, and configuration preferences. Communication via the official MongoDB Node.js driver over TLS. Data model: JSON documents in a 'profiles' collection.
- Vercel REST API / Netlify API: Used to programmatically create and deploy portfolio sites. Input: HTML/CSS/JS build artifacts. Output: stable deployment URL. Authentication via bearer token.
- qrcode (npm): Node.js library used server-side to generate QR code PNG images. Input: portfolio URL string. Output: PNG buffer at 300 DPI with Level H error correction.

3.4 Communications Interfaces

All communications between the ResuFlow frontend, backend, and external APIs use HTTPS (TLS 1.2 or higher) to ensure data confidentiality and integrity. Specific protocols are as follows:

- Frontend to Backend: RESTful HTTP/S API using JSON payloads. File uploads use multipart/form-data encoding.
- Backend to Gemini API: HTTPS POST requests with JSON body. Rate limit: subject to Gemini API quota (requests per minute as per plan tier).
- Backend to MongoDB Atlas: MongoDB Wire Protocol over TLS.
- Backend to Vercel/Netlify API: HTTPS REST calls using JSON and binary (zip) payloads for deployment.
- QR Code delivery to user: Served as a binary PNG over HTTPS from the backend.

4. System Features

The following system features correspond to the major functional components of ResuFlow.

4.1 CV Ingestion and AI Extraction Engine

4.1.1 Description and Priority

This feature handles the intake of user-uploaded CV files and the automated extraction of structured data from them using the Gemini API. This is the foundational feature of the system and is classified as High Priority.

4.1.2 Stimulus/Response Sequences

- User uploads a PDF or DOCX file via the upload screen → System validates file type and size → System extracts raw text → Backend sends text to Gemini API → System receives and validates structured JSON → User is advanced to the Vibe Selection screen.
- User uploads an unsupported file type → System displays an error message: "Unsupported file format. Please upload a PDF or DOCX file."
- File size exceeds 5 MB → System displays an error message: "File too large. The maximum size is 5 MB."
- Gemini API returns an incomplete or invalid JSON profile → System triggers re-extraction with a refined prompt; if still invalid, the user is notified to review their CV.

4.1.3 Functional Requirements

- REQ-1: The system shall accept CV files in PDF and DOCX formats only.
- REQ-2: The system shall reject files larger than 5 MB and display an appropriate error.
- REQ-3: The system shall extract raw text from uploaded files using a server-side parsing library (e.g., pdf-parse for PDF, mammoth for DOCX).
- REQ-4: The system shall transmit extracted raw text to the Gemini API and receive a structured JSON profile in return.
- REQ-5: The JSON profile shall include at minimum: full name, email, phone, summary, skills (hard and soft), work experience (title, company, duration, description), education (degree, institution, year), and achievements.
- REQ-6: The system shall validate the returned JSON against a predefined schema, ensuring required fields, consistent date formats, and standardized field structures.
- REQ-7: The system shall complete the extraction and validation process within 30 seconds for a typical 2-page CV under normal load conditions.

4.2 User Preference and Theme Engine

4.2.1 Description and Priority

This feature enables users to personalize the visual appearance and tone of their portfolio by selecting from predefined options. Priority: High.

4.2.2 Stimulus/Response Sequences

- User selects a color palette → Live portfolio thumbnail updates to reflect selection in real time (< 500 ms).
- The user selects a layout archetype → Portfolio structure updates in the preview panel.
- The user selects an About Me tone → The AI-generated About Me paragraph in the preview reflects the selected tone.
- User clicks "Generate Portfolio" → System maps preferences to design variables and renders the full portfolio.

4.2.3 Functional Requirements

- REQ-8: The system shall offer at least five color palette options (e.g., Classic, Ocean, Earthy, Bold, Monochrome).
- REQ-9: The system shall offer at least three layout archetypes (Minimal, Creative, Corporate).
- REQ-10: The system shall offer at least three tone options for the About Me section (Professional, Casual, Bold).
- REQ-11: Preference selections shall be stored in the user's session and persisted to the MongoDB profile document upon deployment.
- REQ-12: The theme engine shall map user preferences to CSS variables and layout configurations deterministically, ensuring consistent output.

4.3 Portfolio Rendering and Deployment

4.3.1 Description and Priority

This feature handles the generation of the final portfolio HTML/CSS/JS output and its automated deployment to a cloud hosting provider. Priority: High.

4.3.2 Stimulus/Response Sequences

- User clicks "Deploy" on the portfolio preview → System packages portfolio assets → Backend calls Vercel/Netlify deployment API → System polls for deployment status → Upon success, user is redirected to the Dashboard with the live URL.
- Deployment fails → System displays an error with a retry option and logs the failure.

4.3.3 Functional Requirements

- REQ-13: The system shall generate a complete, self-contained HTML/CSS/JS portfolio from the user's JSON profile and selected preferences.
- REQ-14: The system shall programmatically deploy the generated portfolio using the Vercel or Netlify REST API.
- REQ-15: Each deployed portfolio shall be accessible via a unique, stable HTTPS URL.
- REQ-16: Deployment shall complete within 3 minutes from the user initiating the deploy action under normal load.
- REQ-17: The system shall handle deployment API errors gracefully, logging the error and presenting the user with a human-readable failure message and a retry option.

4.4 QR Code Generation

4.4.1 Description and Priority

This feature generates a high-resolution, error-corrected QR code linking to the user's deployed portfolio URL. Priority: Medium.

4.4.2 Stimulus/Response Sequences

- Portfolio is successfully deployed → System automatically generates a QR code → QR code is displayed on the Dashboard screen alongside the portfolio URL.
- User clicks "Download QR Code" → PNG file is downloaded to the user's device.

4.4.3 Functional Requirements

- REQ-18: The system shall generate a QR code immediately upon successful portfolio deployment.
- REQ-19: The QR code shall encode the full HTTPS portfolio URL.
- REQ-20: The QR code shall be generated at a minimum resolution of 300 DPI with Level H error correction (30% data restoration capacity).
- REQ-21: The QR code shall be downloadable as a PNG file.
- REQ-22: The QR code shall remain valid and scannable as long as the deployed portfolio URL is active.

5. Other Non-Functional Requirements

5.1 Performance Requirements

- The AI extraction pipeline (CV text to validated JSON) shall complete within 30 seconds for a standard 2-page CV under normal load (up to 100 concurrent users).
- Portfolio rendering (JSON to HTML/CSS) shall complete within 5 seconds on the server side.
- The frontend shall achieve a Google Lighthouse Performance score of ≥ 85 on desktop.
- Portfolio deployment via Vercel/Netlify API shall complete within 3 minutes under normal platform conditions.
- The system shall support at least 200 concurrent users without degradation of response times beyond 20%.

5.2 Safety Requirements

ResuFlow does not directly control physical systems and therefore has limited safety concerns. However, the following apply:

- User-uploaded CV files shall be scanned for malformed or malicious content before processing (e.g., PDF bomb mitigation by limiting file size and parsing depth).
- The system shall prevent server-side code execution from uploaded file content.

5.3 Security Requirements

- All data transmission between the client and server shall use HTTPS (TLS 1.2+).
- API keys for Gemini, Vercel, and Netlify shall be stored as server-side environment variables and never exposed to the client.
- User CV files and extracted profiles shall be deleted from the server within 24 hours of session expiration, unless the user explicitly opts to save their profile.
- The system shall implement rate limiting on the CV upload endpoint to prevent abuse (maximum 10 uploads per IP per hour).
- User authentication (if implemented) shall use industry-standard protocols (OAuth 2.0 / JWT). For v1.0 without authentication, all portfolio URLs are publicly accessible; no private data is exposed via URLs.

5.4 Software Quality Attributes

- Usability: The end-to-end flow from CV upload to QR code download shall be completable by a non-technical user in under 5 minutes without external guidance.
- Reliability: The system shall achieve 99% uptime (excluding scheduled maintenance and third-party API outages).

- Maintainability: The codebase shall follow modular architecture principles, with each major component (ingestion, AI extraction, theme engine, deployment) encapsulated in separate modules with documented APIs.
- Scalability: The system architecture shall support horizontal scaling of the Node.js backend via containerization (Docker) to accommodate growth beyond 1,000 daily active users.
- Portability: The backend shall be deployable on any cloud provider supporting Docker containers without modification to application code.

5.5 Business Rules

- Each user may have at most one active deployed portfolio URL at a time under a free plan. Re-deploying after preference changes overwrites the previous deployment.
- Portfolio URLs are publicly accessible by anyone with the link or QR code; there is no password protection in v1.0.
- The ResuFlow branding (a small footer badge) shall appear on all deployed portfolios in v1.0. This may be removed in future paid tiers.
- CV data uploaded by users is used solely for portfolio generation and is not used to train AI models or shared with third parties beyond the Gemini API for parsing.

6. Other Requirements

The following additional requirements apply to the ResuFlow system and are not covered in the preceding sections:

- Database: MongoDB Atlas shall be used as the primary data store. The schema shall support versioned user profiles to allow future rollback functionality.
- Internationalization: v1.0 supports English-only CVs. The architecture shall not preclude future addition of multi-language support; all user-facing strings shall be externalized to a localization file.
- Legal: The system must display a Privacy Policy and Terms of Service before a user submits a CV. Users must provide explicit consent for data processing.
- Accessibility: The frontend shall conform to WCAG 2.1 Level AA guidelines. All images shall have alt text; all interactive elements shall be keyboard-navigable.
- Logging: The backend shall implement structured logging (JSON format) for all API requests, extraction events, deployment outcomes, and errors, retained for a minimum of 30 days.

Appendix A: Glossary

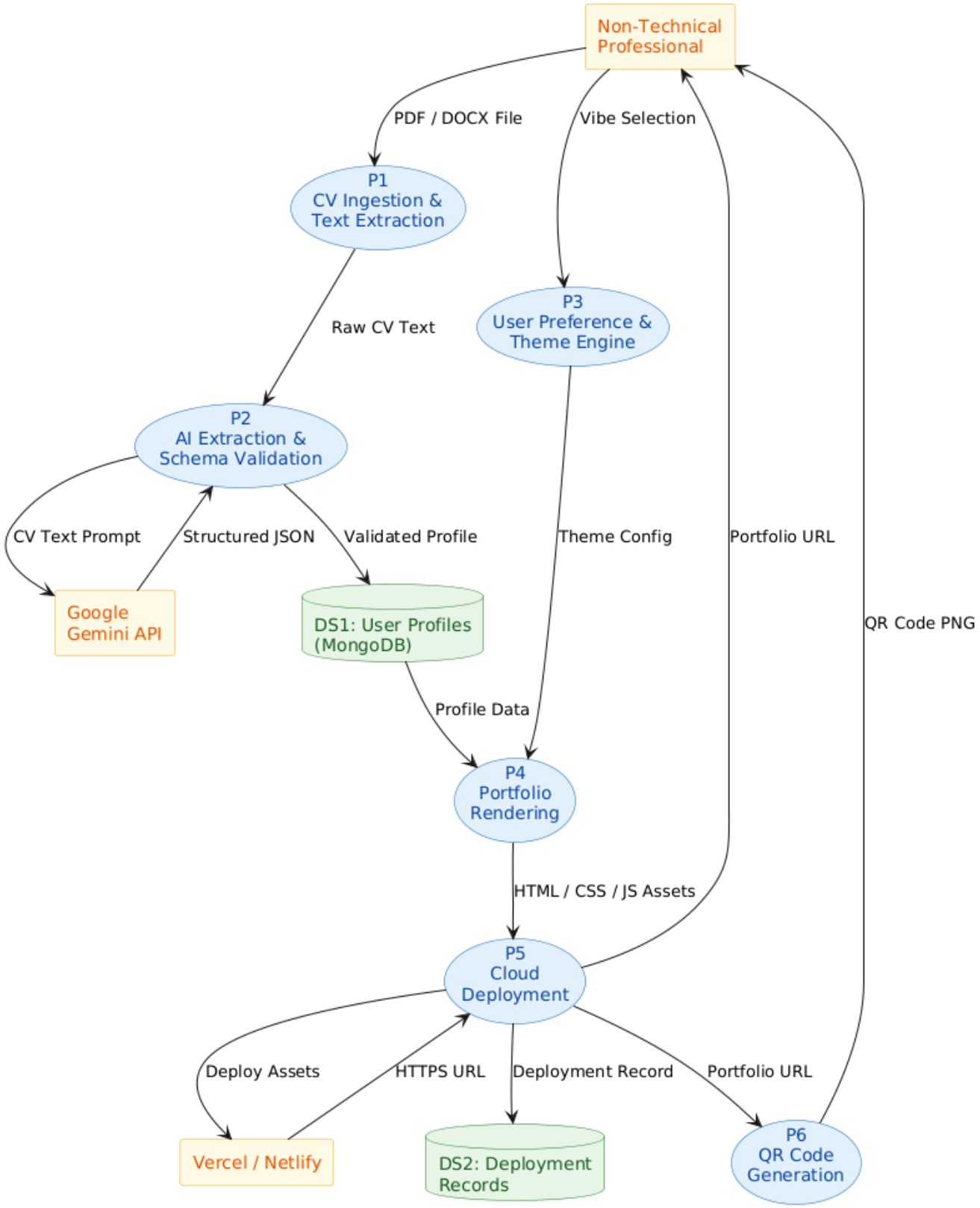
Term	Definition
API	Application Programming Interface — a set of protocols for building and interacting with software applications.
CV	Curriculum Vitae — a document summarizing a person's education, experience, and skills.
DOCX	Microsoft Word Open XML Document format.
Gemini API	Google's generative AI API used for CV parsing and structured data extraction.
JSON	JavaScript Object Notation — a lightweight, human-readable data interchange format.
JWT	JSON Web Token — a compact, self-contained token for secure transmission of information.
PDF	Portable Document Format — a file format for presenting documents independent of software or hardware.
QR Code	Quick Response Code — a two-dimensional barcode that encodes data (e.g., a URL) for scanning.
REST	Representational State Transfer — an architectural style for designing networked APIs.
SPA	Single Page Application — a web application that dynamically rewrites the current page instead of loading new pages.
SRS	Software Requirements Specification — this document.
TBD	To Be Determined — indicates information not yet available.
TLS	Transport Layer Security — a cryptographic protocol for secure communication over networks.
Vibe	ResuFlow's user-friendly term for the combination of color palette, layout archetype, and tone that defines a portfolio's aesthetic.

Appendix B: Analysis Models

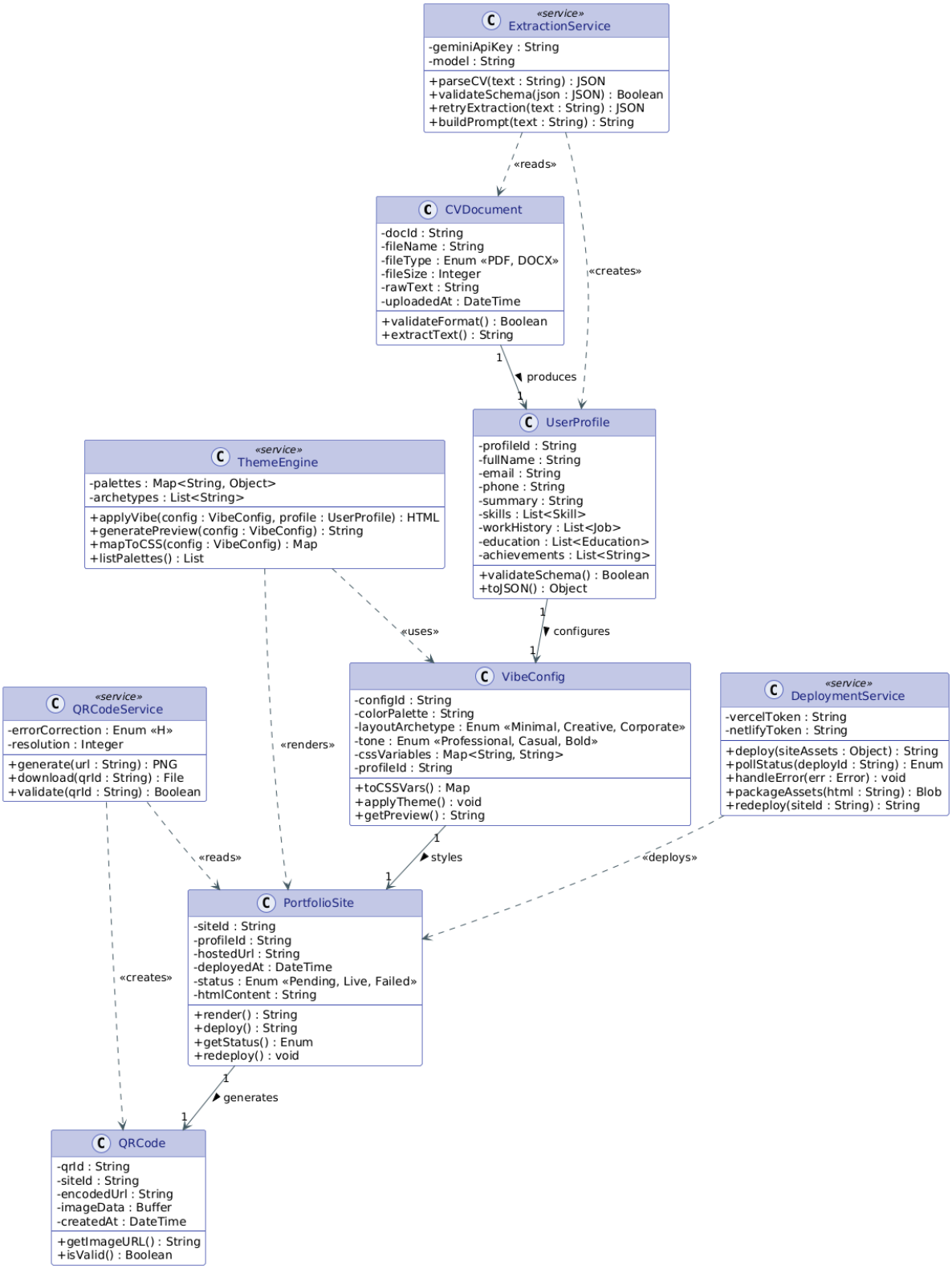
The following analysis models are referenced for the ResuFlow system. Detailed diagrams are to be included in a supplementary design document (Version 1.0 — TBD):

- System Context Diagram: Illustrates ResuFlow's interactions with external entities (User, Gemini API, MongoDB Atlas, Vercel/Netlify API).
- Data Flow Diagram (Level 0): Shows the top-level data flows: CV Input → AI Extraction → Profile Storage → Theme Engine → Portfolio Deployment.
- Entity-Relationship Diagram: Defines the MongoDB data model, including the User Profile document structure and its relationships to session data and deployment records.
- Use Case Diagram: Depicts the primary use cases for the Non-Technical Professional user class: Upload CV, Configure Vibe, Preview Portfolio, Deploy Portfolio, Download QR Code.

DFD Level 1


Non-Technical Professional
PDF / DOCX File
Vibe Selection
P1 CV Ingestion & Text Extraction
P3 User Preference & Theme Engine
Raw CV Text
P2 AI Extraction & Schema Validation
CV Text Prompt
Structured JSON
Validated Profile
Theme Config
Portfolio URL
Google Gemini API
DS1: User Profiles (MongoDB)
QR Code PNG
Profile Data
P4 Portfolio Rendering
HTML / CSS / JS Assets
P5 Cloud Deployment
Deploy Assets
HTTPS URL
Deployment Record
Portfolio URL
Vercel / Netlify
DS2: Deployment Records
P6 QR Code Generation

Class Diagram



Appendix C : To Be Determined List

- TBD-1 (Section 2.4): Finalize decision between Vercel and Netlify as the primary deployment target based on API capability comparison.
- TBD-2 (Section 4.2): Define the complete set of color palettes and layout archetypes with design mockups.
- TBD-3 (Section 4.3): Establish specific Vercel/Netlify API endpoints and authentication flow for automated deployment.
- TBD-4 (Section 5.3): Determine authentication strategy for v1.0 (session-based anonymous access vs. lightweight account creation).
- TBD-5 (Section 6): Draft Privacy Policy and Terms of Service documents for legal review.